CURRENT ACQUISITION UNIT (CAU)

1.1 Features

- The A89201 has a dedicated current acquisition unit for motor control applications
- Three SAR (Successive Approximation Register) analog-to-digital converters.
- 12-bit resolution conversions
- Configurable track-and-hold (T/H) trigger sources
- Configurable track -and-hold (T/H) periods
- Configurable conversion speed (up to 1 µs)
- Zero-current offset calibration functionality
- Automatic calibration functionality
- Optional IIR filtering of zero-current calibration signals
- Comprehensive interrupt sources and error flags with interrupt enables for conversion completion and error detection functionalities
- DMA support

1.2 Introduction

The A89201 features 3 analog-to-digital converters (ADC) dedicated entirely to current acquisition in three-phase motor control applications. The 3 ADC channels are characterized by their fast conversion times, which can be as fast as 1 µs, while maintaining 12-bit resolution for accurate current measurements and smooth control. Each ADC channel is configured with high flexibiltiy to select the desired conversion speed, track and hold period, conversion trigger source and trigger signal polarity. The module also features an advanced automatic calibration functionality, which takes care of any voltage offsets caused by the sense amplifiers. The calibration system features an optional IIR filter to remove any unwanted noise during the calibration process.

1.3 Functional Description

The CAU module hosts 3 ADC channels, dedicated for current acquisition in three-phase systems. Each ADC channel is connected to the sense amplifier output. More detailed information about the sense amplifier electrical characteristic is described in the GDU section. Each ADC channel can be configured separately through its CAU_ADCx register. The MCU selects the trigger source to start an ADC conversion from the CAU_TRIGx register. 14 trigger sources are available for each ADC channel. These sources include trigger signals generated by the pulse width modulation unit (PWMU), triggers generated by software, external signals connected through the GPIO module or signals generated by the general purpose timer unit (GTU). Once the conversion is complete, the result of each ADC channel is stored in its corresponding CAU_RESULTx register. The module features a calibration system that is used to calculate the zero-current voltage offset resulting from the differential sense amplifiers, which measure the voltage across the shunt resistors. The calibration trigger source is selected similar to the normal triggers. The resulting offset is stored in the CAU_OFFSETx register. The measured offset voltage can be filtered with an IIR filter to remove any unwanted noise or discripancies in the signal. The automatic calibration function will calculate the result of any conversion with respect to the detected offset. The module generates an interrupt request if any conversion errors are detected or when a conversion is complete. Lastly, the module features a DMA interface controller and DMA handshake signals to support DMA functionality. Figure 1-1 shows the general block diagram of CAU module. The same structure is repeated for the remaining ADC channels.

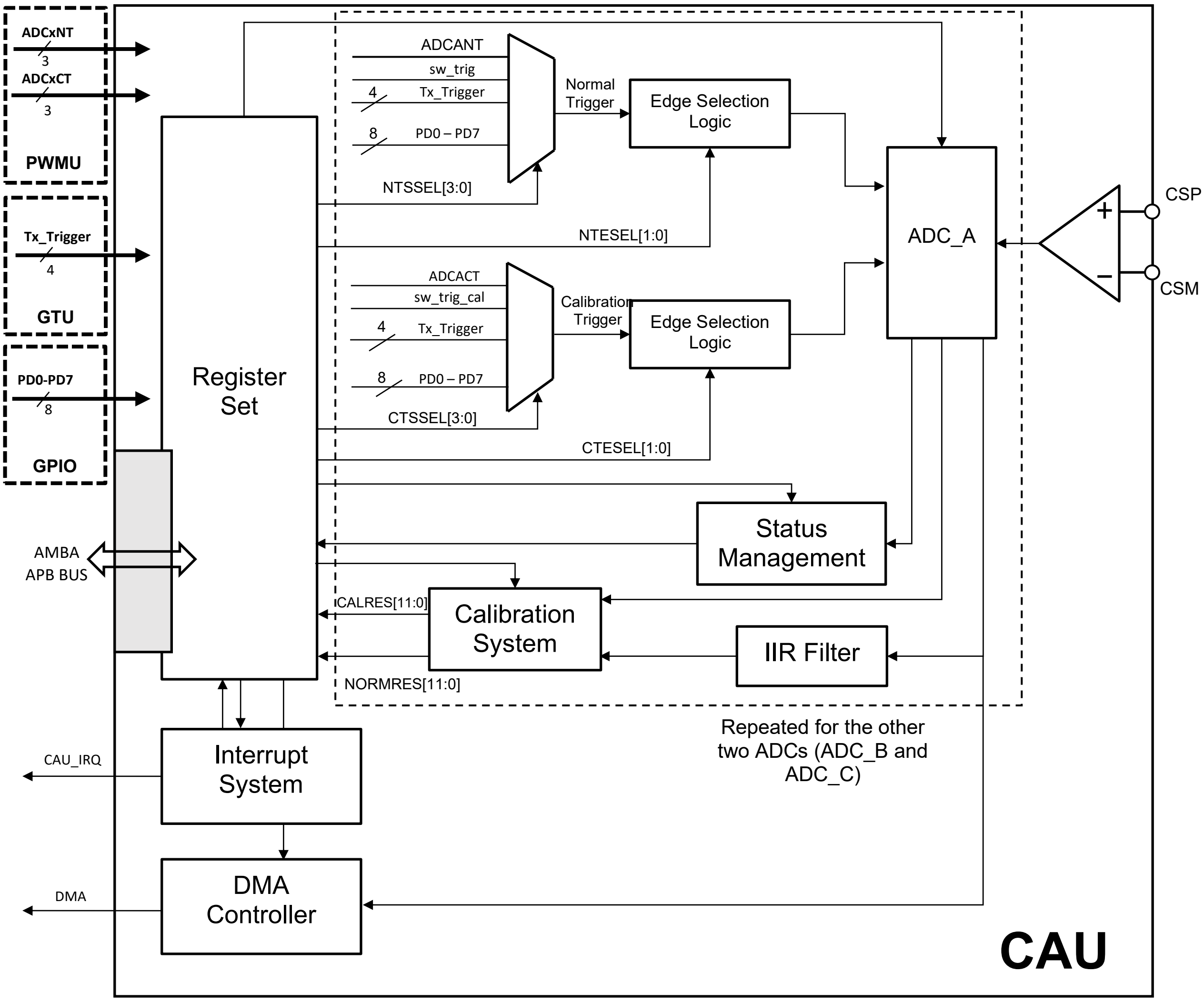


Figure 1-1: Block diagram of the CAU block

1.4 Operational Description

1.4.1.1 ADC Offset Measurement

System offset is caused by the current sense amplifiers. To counter this offset, the amplifier output voltage should be measured at zero-current (this is controlled by the calibration trigger signal at a right time) to remove the offset. The calibration system features an IIR filter that aims at removing unwanted noise during offset calculation. The output raw voltage is therefore filtered through the IIR filter, if the option is enabled from the CAU_IIRx registers, and the output of the filter is considered as the offset value. This offset value (CALRES[1]) is stored in the CAU_OFFSETx register and is used to calibrate future conversions automatically.

The offset is calculated as follows:

$$\text{CALRES} = A' + (N - N')$$

Where A is the new raw value coming from the ADC after a calibration trigger. N is the previous offset value. A' and N' are the signals A and N respectively divided by the filter coefficients 2^n where n is selected by the CF_COE bits in the CAU_IIRx register.

$$A' = \frac{A}{2^n} \quad ; \quad N' = \frac{N}{2^n}$$

The result of the offset is a 12-bit signed number. The relationship between CALRES and calibration voltage across sense resistor can be calculated as below eventually:

$$V_{RSENSECAL}(V) = \frac{\textbf{CALRES}}{2048} \times \frac{1.2}{SAG}$$

Where SAG is the sense amplifier gain that can modified by register 1 in GDU (refer to GDU documentation for further information).

For example when SAG is set to ‘5’, CALRES[11:0] value set to ‘40’, then the voltage across sense resistor can be calculated as below:

$$V_{RSENSECAL} = \frac{40}{2048} \times \frac{1.2}{5} = 4.69mV$$

Note: (1) To obtain CALRES value, read the corresponding CAU_OFFSETx register as a signed 16-bit number and shift that value by two bit to the right.

1.4.1.2 ADC Offset Removal

Depending on the automatic calibration enable bit, ACM (within CAU_ADCx register), the CAU will provide a non-calibrated or a calibrated result and it is stored as 12bit signed number NORMRES in the CAU_RESULTx. If the calibration feature is selected, the latest offset value (CALRES) obtained in a calibration cycle in the CAU_OFFSETx register is subtracted from the uncalibrated result (the result coming from an ADC after a normal trigger) and NORMRES in the CAU_RESULTx register will be updated according to below equation:

$$\mathrm{NORMRES} = \mathrm{uncal\_result} - \mathrm{CALRES}$$

NORMRES is clamped to ±2048 to avoid over-flow or underflow after offset removal.

If the calibration feature is not selected, the uncalibrated result shows the direct measurment acorss the sense resistor after a normal trigger:

$$\mathrm{NORMRES} = \mathrm{uncal\_result}$$

Where uncal_result is the uncalibrated measurement coming from the ADC after a normal trigger.

Notes:

(1) The result of calibration will be saved in CAU_OFFSETx register and if automatic calibration is not desired, this value can be manually subtracted from CAU_RESULTx register. To turn off automatic calibration ACM in CAU_ADCx needs to be set to 0.

(2) x represents the corresponding ADC Channel (ADC_A, ADC_B, ADC_C)

1.4.2 ADC Conversion Calculation

The CAU logic provides each ADC channel with signals to start the conversion and waits for the result. To start a new conversion, firstly, the ADC must be enabled in the CAU_ADCx register. The T/H trigger source must be selected from the CAU_TRIGx register. Once a track-and-hold trigger is detected (depending on the selected edge detection settings from the CAU_TRIGx register), the conversion is started. The analog circuit will sample the data after a period defined by the CASH_PRD information defined in the CAU_TRIGx register. The conversion speed is defined by the CACNV_SPEED field in the CAU_ADCx register. Once the conversion is completed, the NORMRES[1] value is updated in CAU_RESULTx[2] register with a new value. The NORMRES result is a 12-bit signed number. The analog to digital conversion across the sense resistor is calculated according to below equation:

$$V_{RSENSE}(V) = \frac{\text{NORMRES}}{2048} \times \frac{1.2}{SAG}$$

Where SAG is the sense amplifier gain that can modified by register 1 in GDU (refer to GDU documentation for further information).

For example when SAG is set to ‘5’, and the NORMRES value set to ‘-2000’, then the voltage across sense resistor can be calculated as below:

$$V_{RSENSE} = \frac{-2000}{2048} \times \frac{1.2}{5} = -234\,mV$$

Note: (1) To obtain NORMRES value, read the corresponding CAU_RESULTx register as a signed 16-bit number and shift that value by two bit to the right.

(2) x represents the corresponding ADC Channel (ADC_A, ADC_B, ADC_C)

1.4.3 Trigger selection

There are two types of triggers in the CAU module: normal triggers and calibration triggers. Normal triggers , are used to start a conversion and the resulting conversion is stored in the CAU_RESULTx register. Calibration triggers are used to start a calibration cycle (or offset calculation) and the resulting conversion is stored in the CAU_OFFSETx register. Normal triggers have a higher priority over calibration triggers.

1.4.4 Normal Trigger selection and settings

The CAU normal triggers can be generated by the PWM block, ADCxNT, software triggers, Tx_Trigger signals generated by the GTU block or by external trigger signals imported through the GPIO pins. The trigger source is selected by the NTSSEL field in the CAU_TRIGx registers. The edge detection selection is also done by the same register using the NTESEL field. Finally, the field CATH_PRD determines the track and hold period.To use software triggers, SW_TRIG must be selected in the trigger source field (NTSSEL = 0001_B) in the CAU_TRIGx register. The software trigger is controlled by the CAU_SWTRIG register.

1.4.5 Calibration Trigger selection and setting

The CAU features a calibration and filtering system that is used to remove any offsets and unwanted noise in the system. There are several calibration trigger sources: calibration triggers, ADCxCT, generated from the PWM module, a calibration specific software trigger. The Tx_Trigger signals from GTU and external signals connected through GPIO. The calibration trigger source and edge detection selection can be done by fields CTSSEL and CTESEL respectively in CAU_CALIBRATIONx register. T/H duration and conversion speed are adopted from the configurations set in the CAU_TRIGx and CAU_ADCx registers respectively. ie, these configurations are used for both normal and calibration triggers. When calibration trigger source, CTSSEL , is set to SW_TRIG_CAL, then the trigger source is generated by changing the paratmerrs in CAU_SWTRIG_CAL register in the software.

1.4.6 Status, over-run and overflow detection

The CAU_ERR_STATUS register provides diagnostic information of the CAU module’s status and any occuring faults. It provides information about status, over-run and out-of-range detection. The status register also shows which ADC’s conversion has failed and why.

The over-run bits, OVR, indicate whether an over-run error is detected. An over-run error occurs when one of the result registers is overwritten before being read out by the host controller. The OVR bits indicate the over-run detection for each ADC channel. The overwrite functionality is configured by the OW_SEL bit in the CAU_ADCx register for each ADC channel. If the overwrite function is not permitted, the CAU_RESULTx register is not updated with the new value and the OVR bits are set to indicate an error. If the overwrite function is permitted, the CAU_RESULTx register is overwritten with the new value and the OVR bits will be still set to notify the user of the overwrite event.

The out-of-range, OOR, bits indicate whether an out of range error (overflow or underflow) is detected. This error is caused during the calibration between the non-calibrated result and the offset value. If the calibrated result, after the subtraction of the offset value, from the non-calibrated result, is out of range causing either an underflow or overflow, the OOR bits are set indicating which ADC channel experienced the error. The result value is set to the maximum value within the range when an overflow is detected, and set to the minimum value within the range when an underflow is detected.

The CTRIG_SKIP bits indicate when a normal trigger has been ignored by one of the ADC channels. This might be the case if the ADC is not ready to start a new conversion and receives a new trigger. An example of such situation can happen when the CAU is enabled and a conversion is ongoing or when the CAU is enabled but the ADC is still in the power-up period.

The TRIG_PWRUP bit indicates that a trigger was received when the ADC was still powering up. This bit is flagged in conjunction with the appropraite CTRIG_SKIP bit. From the information in both these fields, the ADC channel that skipped the trigger can be determined.

The RCONF_ERR bits indicate an invalid attempt to modify the CAU_ADCx, CAU_TRIGx or CAU_CALIBRATIONx registers' contents while a conversion is ongoing. Modifying the register configuration will be accepted, however, these changes will not take effect until the next conversion is triggered. In other words, the ongoing conversion will use the old configuration and the following conversion will use the updated configuration. This error flag is set to notify the user of an attempt to change the ADC configuration during a conversion. An exception to this is disabling an ADC channel via the enable bit (ADC_EN) in the CAU_ADCx registers. If the ADC is disabled by software, even if a conversion is ongoing, it will be disabled instantly and the ongoing conversion will be aborted. The RCONF_ERR flag will be set even though the ADC is disabled.

The TRIG_ERR bits indicate that a normal trigger and a calibration trigger have been initiated for a given ADC channel at the same time. In this case, the normal trigger is priotirized and the calibration trigger is ignored.

1.4.7 ADC power up

When an ADC channel is enabled, it enters its power-up period. The power-up period lasts for 200 clock cycles. During this period, the ADC will not accept any triggers (normal or calibration). If a trigger is received within this period, the respective CTRIG_SKIP bit is set in the CAU_ERR_STATUS register along with the TRIG_PWRUP bit to indicate that an invalid trigger is received during power up.

To monitor the state of the ADC channels, the CAU_PWRUP_STATUS register provides information regarding whether each ADC is ready or not to accept a trigger. Each bit corresponds to one ADC channel. If the bit displays 0, this means that the ADC is either disabled or still in its power-up period. Meaning that this ADC channel is not ready to accept any triggers. If the bit displays 1, this means that the ADC channel is fully powered up and ready to receive triggers and start a conversion.

This information describes the readiness of each ADC channel in respect to its power-up, and not to be confused with whether the ADC is busy completing another conversion from a previously received trigger. The CAU_CONV_STATUS provides information to check whether an older conversion has been completed or not.

1.4.8 Interrupt generation and management

A single interrupt request (IRQ) signal is generated by the CAU module to the MCU's Nested Vectored Interrupt Controller (NVIC). The module signals an interrupt request to report the completion of an ADC conversion or the detection of an error. The generated IRQ signal is an active high signal.

The interrupt management system takes information from the CAU_CONTROL, CAU_CONV_INTEN, CAU_ERR_INTEN, CAU_CONV_STATUS and CAU_ERR_STATUS registers. Depending on this information, the interrupt management system makes the decision of whether to generate an interrupt request or not.

The information stored in the CAU_CONV_INTEN register defines whether an interrupt generation is enabled or disabled on the completion of a conversion for each ADC channel.

The information stored in the CAU_ERR_INTEN register defines whether an interrupt generation is enabled or disabled on each error detected for each ADC channel except for the TRIG_PWRUP_EN signal which is common between all 3 ADC channels.

The information stored in the CAU_CONV_STATUS register shows whether a conversion is complete or not for each ADC channel. Writing '1' to each field will reset the flag. Once any of these bits are set, an IRQ is generated to the host's NVIC if the interrupt generation is enabled via the CAU_CONV_INTEN register. If the interrupt generation is disabled, the CAU_CONV_STATUS register will still be updated to report the completion of any conversion.

The information stored in the CAU_ERR_STATUS register shows if any error has occurred on any of the ADC channels. Once any of these bits is set, an IRQ is generated to the host's NVIC if the interrupt generation for the corresponding error is enabled via the CAU_ERR_INTEN register. If the interrupt generation is disabled, the CAU_ERR_STATUS will still be updated to report any errors without interrupting the host. Writing '1' to each field will reset the flag.

To generate a single IRQ signal, the error detection information is OR-ed with the conversion completion information. Therefore, upon receiving the an interrupt request, the user software must investigate the source of the interrupt to identify whether it was caused due to a conversion completion or the detection of an error.

Note: It is more likely that a conversion complete occurs than an error in conversion. CAU_CONV_STATUS register should be checked first.

When CAU is disabled, CAU_ERR_STATUS and CAU_CONV_STATUS registers will be reset. Internal logic will be set to the default values.when CAU enabled, CAU_CONV_STATUS, and CAU_ERR_STATUS registers can be cleared at any time by writing '1' to the asserted field. The interrupt-generation configuration registers CAU_CONV_INTEN and CAU_ERR_INTEN will retain their values if the CAU module is disabled to keep their configured settings when the CAU is enabled once again. If a reset occurs, (eg. POR), these registers will be reset to their default values.

Figure 1-2 shows the interrupt logic of the CAU module. Figure 1-3 shows the error managements logic.

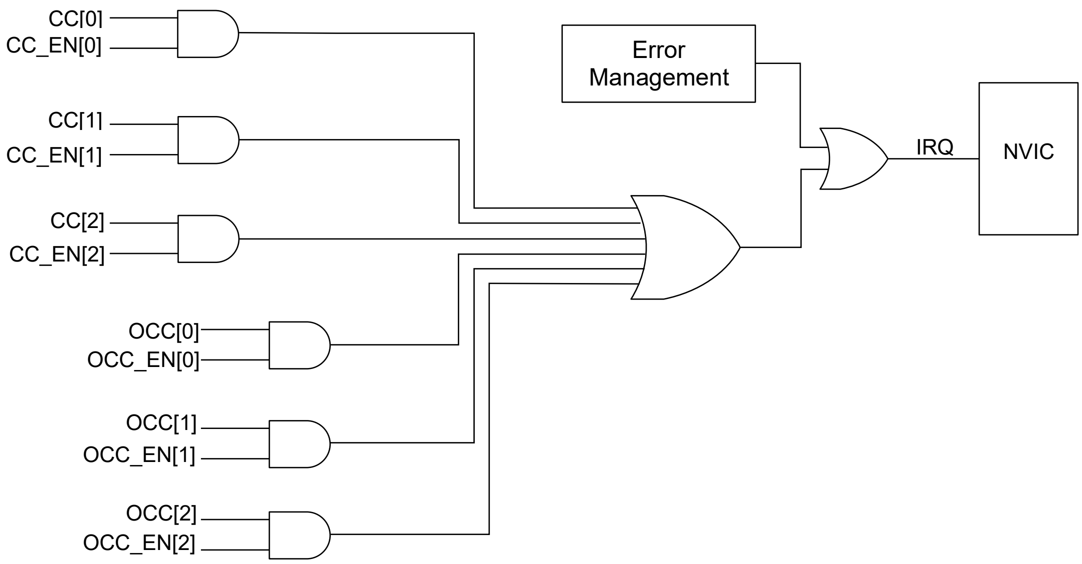


Figure 1-2: Interrupt control logic

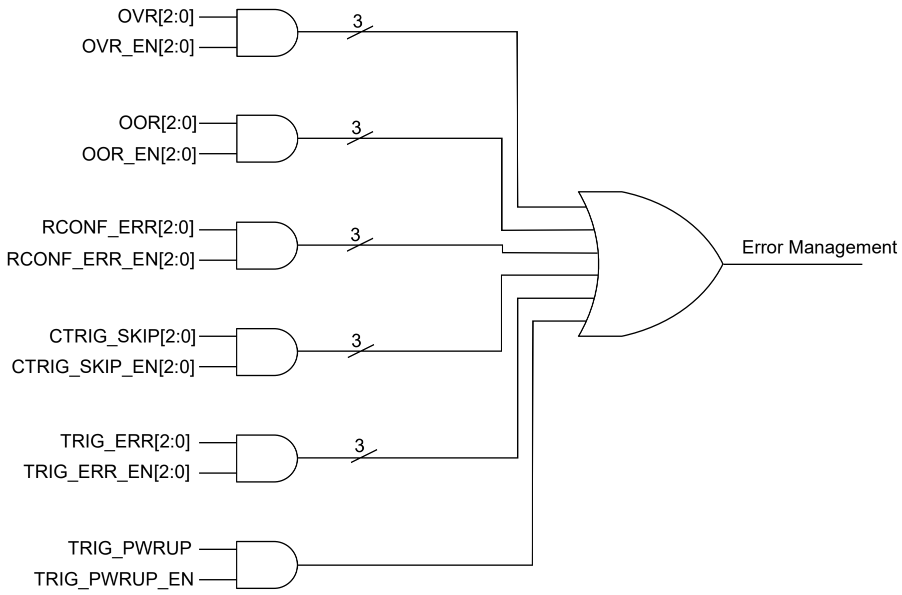


Figure 1-3: Error management logic

1.4.9 CAU and ADC enable

The CAU module features a general enable switch to enable and disable the entire CAU module, as well as three ADC specific enable switches. The CAU enable bit is found in the CAU_CONTROL register. Each ADC has its own enable bit found in its corresponding CAU_ADCx register. This section describes the behavior of the module upon disabling each of the mentioned enable switches.

1.4.9.1 Disabling the CAU module

Upon disabling the CAU module, via the CAU_CONTROL register, all ADC channels are automatically disabled regardless of the value of the enable bit in each of the CAU_ADCx registers. However, in this case, the ADC_EN values in each CAU_ADCx registers maintain their values. Such that their current configuration will be restored once the CAU module is enabled once again. If the CAU is reset, the values are lost and the default values are assumed.

Disabling the CAU will reset the CAU_ERR_STATUS, CAU_CONV_STATUS, CAU_OFFSETx and CAU_RESULTx registers. The interreupt enable registers CAU_ERR_INTEN and CAU_CONV_INTEN registers will retain their values such that their configuration is restored once the CAU is enabled once again.

Any issued irq will be reset once the CAU module is disabled. Since both the CAU_ERR_STATUS and CAU_CONV_STATUS registers will be reset by disabling the CAU module, an IRQ will not be triggered once the CAU module is enabled once again.

1.4.9.2 Disabling the ADC channels

Each ADC channel can be enabled or disabled independently of the other two channels via the ADC_EN field in the corresponding CAU_ADCx register. Disabling a given ADC channel will reset its corresponding bits in the CAU_ERR_STATUS and CAU_CONV_STATUS registers. The corresponding CAU_OFFSETx and CAU_RESULTx will be also reset. If an IRQ was set by a given ADC, it will be automatically reset if the ADC is disabled.

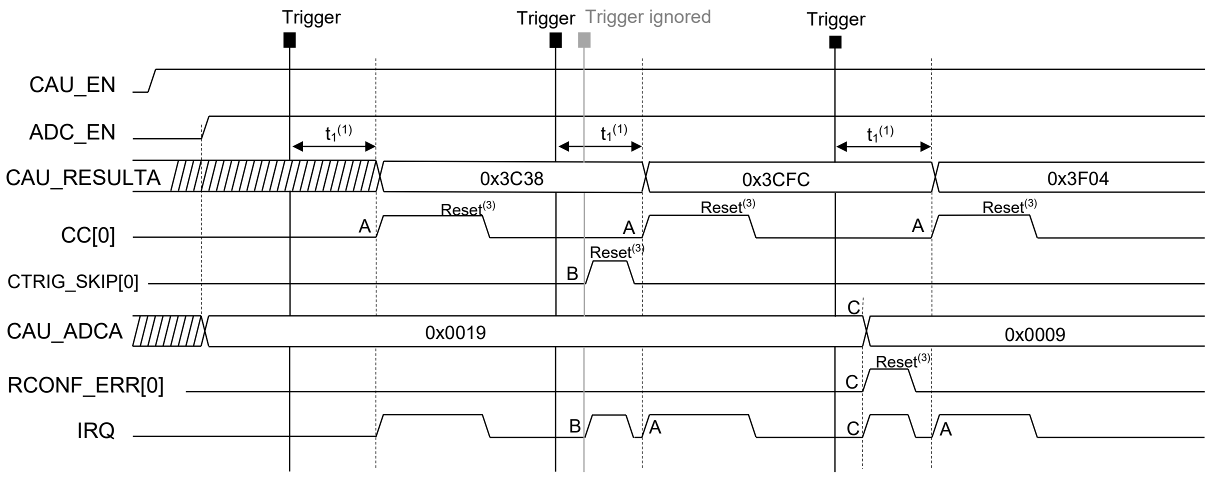
Enabling any of the ADC channels will have no effect until the CAU module is enabled.

1.5 Behavioural examples

In this section, operational examples are shown to illustrate the behavior of the CAU module. Each example will incorporate different scenarios to show how the CAU would behave in each situation.

1.5.1 Example 1

In this example ADC_A is enabled and configured by writing to the CAU_ADCA register. The T/H trigger settings are configured via the CAU_TRIGA register (not shown in this example). Once the T/H trigger is detected, the conversion will start automatically. Once the conversion is complete, the CAU will set the first bit in conversion complete, CC, field in the CAU_CONV_STATUS register to indicate the completion of the conversion, shown in the figure below as event A. Depending on the interrupt generation settings, configured via the CAU_CONV_INTEN register, an IRQ will be generated to the system NVIC. Figure 1-4 shows a sequence of events and how the CAU would handle each sequence for ADCA. The same rule applies to the other two ADCs , ADCB and ADCC. This example shows how the module reacts if a second trigger is received before the initial conversion is completed, shown in the figure as event B. The example also shows the behavior of the module if the CAU_ADCA register (same applies for CAU_CALIBRATIONA and CAU_TRIGA registers) configuration is changed while a conversion is ongoing, shown as event C. The register contents will be updated but the changes will not take effect before the current conversion is complete.



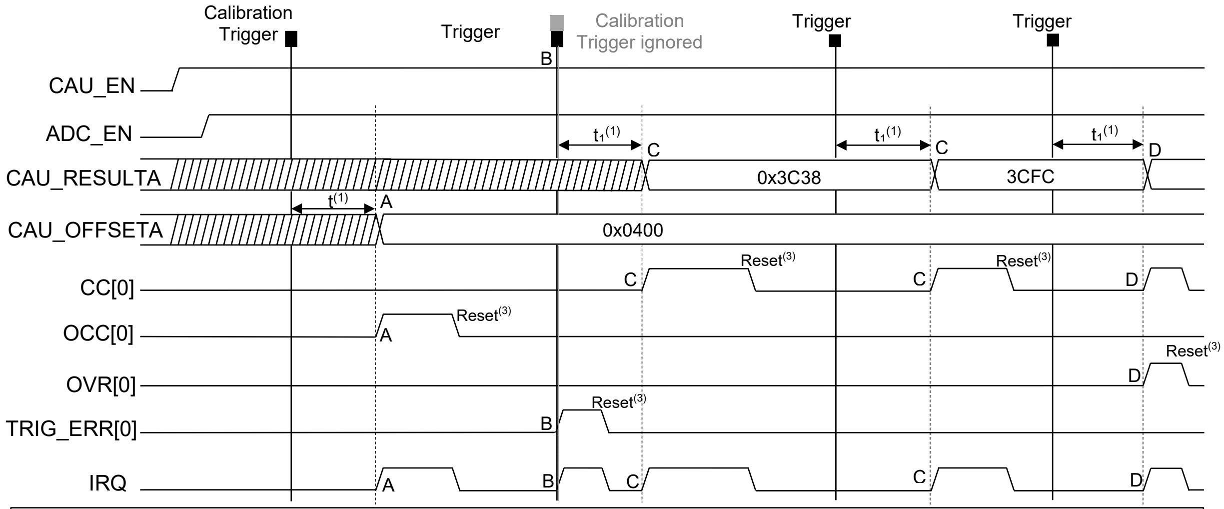
- **Event A:** ADCA conversion is completed. The conversion result is stored in the CAU_RESULTA register. The flag CC[0] in the CAU_CONV_STATUS register is set, triggering an IRQ(2) to the NVIC.
- **Event B:** ADCA receives a trigger while a conversion is ongoing. The fault flag CTRIG_SKIP is set in the CAU_ERR_STATUS register, triggering an IRQ(2) to the NVIC.
- **Event C:** ADCA configuration register (CAU_ADCA) is changed while a conversion is ongoing. The register contents are modified in the register map, but changes will take effect only when the current conversion is done(4), that is by the following point A. The next conversion will use the updated configuration. Event C will set the RCONF_ERR fault flag in the CAU_ERR_STATUS register which will trigger the IRQ(2) to the NVIC.

Notes:
(1) t_1 = T/H time + Conversion time as defined in the CAU_TRIGA and CAU_ADCA registers respectively.
(2) IRQ is triggered assuming that irq generation is enabled in CAU_ERR_INTEN and CAU_CONV_INTEN registers.
(3) Signal is reset by software by writing '1' to CAU_CONV_STATUS or CAU_ERR_STATUS registers. The signal IRQ follows these signals and hence will be reset too.
(4) An exception to this is the ADC_EN signal in the CAU_ADCx registers. If this field is toggled, the effect will take place right away without waiting for the current conversion to complete.

Figure 1-4: CAU behaviour – example 1

1.5.2 Example 2

In this example ADC_A is enabled. Once the calibration trigger is detected, the conversion will start automatically. Once the conversion is complete, the CAU will set the OCC[0] bit in the CAU_CONV_STATUS register to indicate the completion of the calibration conversion, shown in the figure as event A. The resulting conversion will be stored in the CAU_OFFSETA register. Depending on the interrupt generation settings, configured via the CAU_CONV_INTEN register, an IRQ will be generated to the system NVIC. Figure 1-5 shows a sequence of events and how the CAU would handle each sequence. This example shows how the module reacts if a calibration and a normal trigger are received at the same time, shown in the figure as event B. The example also shows the behavior of the module if a conversion is complete before the previous result is read out from the CAU_RESULTA register. In this example, the module is configured to overwrite the CAU_RESULTA register with the new value. The OVR flag is set to indicate that an overrun fault has occurred which will trigger an IRQ. This scenario is shown as event D.



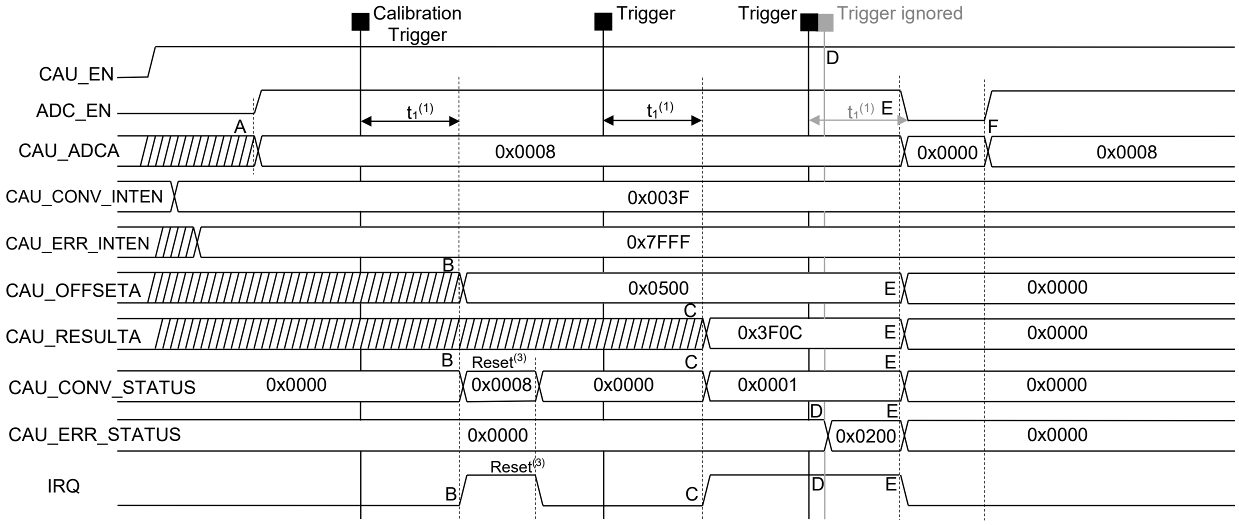
- **Event A:** ADC_A calibration conversion, triggered by a calibration trigger, is completed. The conversion result is stored in the CAU_OFFSETA register. T/H period and conversion time are adopted from the normal trigger configuration in CAU_TRIGA and CAU_ADCA registers. The flag OCC[0] in the CAU_CONV_STATUS register is set, triggering an IRQ(2) to the NVIC.
- **Event B:** ADC_A receives a normal trigger and a calibration trigger at the same time. In this case, the normal trigger is prioritized, and a normal conversion starts. The fault flag TRIG_ERR[0] is set in the CAU_ERR_STATUS register, triggering an IRQ(2) to the NVIC, to indicate that a calibration trigger has been ignored.
- **Event C:** ADC_A normal conversion is complete. The conversion result is stored in the CAU_RESULTA register. The flag CC[0] in the CAU_CONV_STATUS register is set, triggering an IRQ(2) to the NVIC.
- **Event D:** A normal trigger is received, and the new conversion is complete before the previous result has been read out from the CAU_RESULTA register. The CAU_RESULTA register is updated with the new value if the overwrite option is permitted via the OW_SEL bits in the CAU_ADCA register. The OVR[0] fault flag is set to indicate that the old conversion value has been overwritten without being read out. In case the overwrite option is not permitted, the CAU_RESULTA register is not updated with the new conversion. The OVR[0] fault flag is set to indicate that a new conversion is missed and the result register is not updated. The fault flag will in turn trigger the IRQ(2) to the NVIC.

Notes:
(1) t_1 = T/H time + Conversion time as defined in the CAU_TRIGA and CAU_ADCA registers respectively.
(2) IRQ is triggered assuming that irq generation is enabled in CAU_ERR_INTEN and CAU_CONV_INTEN registers.
(3) Signal is reset by software by writing '1' to CAU_CONV_STATUS or CAU_ERR_STATUS registers. The signal IRQ follows these signals and hence will be reset too.

Figure 1-5: CAU behaviour – example 2

1.5.3 Example 3

This example shows the CAU's behavior when an ADC channel is disabled. When an ADC channel is disabled, all bits corresponding to this channel in the CAU_CONV_STATUS or CAU_ERR_STATUS registers are reset. The corresponding CAU_RESULTx and CAU_OFFSETx registers are also reset. Configuration registers such CAU_ADCx, CAU_TRIGx, CAU_CALIBRATIONx, CAU_CONV_INTEN and CAU_ERR_INTEN registers remain unchanged unless the entire module is reset (eg. POR). This is to maintain the configuration in case the ADC channel is re-enabled later on. In this example, ADC_A is enabled. The module is configured to generate interrupts for any conversion completion or any errors occuring via the CAU_CONV_INTEN and CAU_ERR_INTEN registers A calibration trigger is detected and thus the offset is calculated and an IRQ is generated. A normal trigger is then detected and a normal conversion is completed. Two normal triggers are then detected. The second trigger is detected before the first conversion is completed. This will generate an error and the flag CTRIG_SKIP is set, generating an IRQ. At point E, the ADC is disabled (while a conversion is on-going), The conversion is aborted, CAU_OFFSETA, CAU_RESULTA are reset. Any error bits or transmission completion bits in the CAU_ERR_STATUS and CAU_CONV_STATUS registers will be reset as well. Configuration registers such as CAU_CONV_INTEN and CAU_ERR_INTEN retain their values. Figure 1-6 shows the behavior of the CAU module in this example.



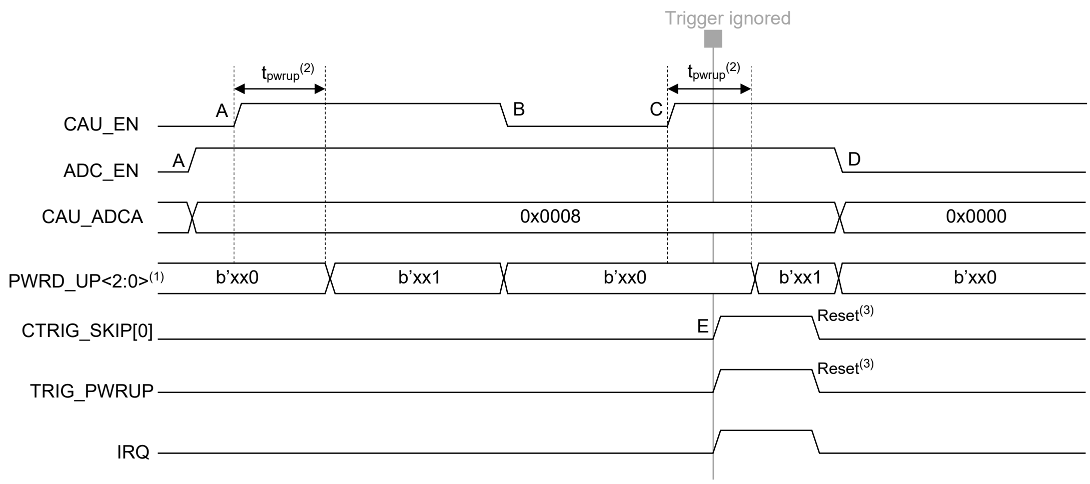
- **Event A:** ADC_A is enabled by setting the ADC_EN bit in the CAU_ADCA register after enabling the CAU and writing the configuration registers CAU_CONV_INTEN, CAU_ERR_INTEN, CAU_TRIGA and CAU_CALIBRATIONA.
- **Event B:** A calibration trigger was detected and thus the offset has been calculated. The offset value is stored in the CAU_OFFSETA register and the OCC[0] is set in CAU_CONV_STATUS register to indicate the successful completion of this conversion, triggering an IRQ(2).
- **Event C:** A normal trigger was detected, and a normal conversion has commenced. The resulting value is stored in the CAU_RESULTA register and the CC[0] in CAU_CONV_STATUS register is set to indicate the successful completion of a conversion, triggering an IRQ(2).
- **Event D:** Two consecutive normal triggers are received (before the first conversion is completed). The second trigger is then ignored and the CTRIG_SKIP fault flag is set in the CAU_ERR_STATUS register. This will trigger an IRQ(2,4).
- **Event E:** At this point, ADC_A is disabled by resetting the ADC_EN field in the CAU_ADCA register. Note that a conversion was ongoing, and this would typically flag a RCONF_ERR fault. However, disabling the ADC will take effect immediately. The ongoing conversion is aborted, CAU_OFFSETA and CAU_RESULTA registers are reset. Any fault flags or conversion completion flags related to ADC_A in the CAU_ERR_STATUS and CAU_CONV_STATUS registers, respectively, are reset as well (faults and completion flags of other ADC channels are not altered). Consequently, any IRQ generated by the ADC_A will be reset.
- **Event F:** Upon setting ADC_EN, the CAU_ERR_INTEN, CAU_CONV_INTEN, CAU_CALIBRATIONA, CAU_TRIGA registers are unchanged and retain their previous values.

Notes:
(1) t_1 = T/H time + Conversion time as defined in the CAU_TRIGA and CAU_ADCA registers respectively.
(2) IRQ is triggered assuming that irq generation is enabled in CAU_ERR_INTEN and CAU_CONV_INTEN registers.
(3) Signal is reset by software by writing '1' to CAU_CONV_STATUS or CAU_ERR_STATUS registers. The signal IRQ follows these signals and hence will be reset too.
(4) The IRQ signal in this case is assumed to not be reset by software for illustration purposes.

Figure 1-6: CAU behaviour – example 3

1.5.4 Example 4

In this example, the ADC power-up sequence is demonstrated in Figure 1-7. The ADC is enabled when the ADC_EN signal is high and CAU is enabled. Changing ADC_EN will not have any effect if CAU is not enabled. On the other hand, disabling CAU will not affect the setting of ADC_EN. In other words, the CAU_ADCx register retains its value so the ADC will restore its configuration right away when CAU is enabled once again. If a trigger (normal or calibration) is received when the ADC is not enabled (or still in its power-up period), the trigger will be skipped and the CTRIG_SKIPX and TRIG_PWRUP signals will be set to indicate that a trigger has been received when the ADC was not powered up yet. If interrupt generation for these two signals is enabled from the CAU_ERR_INTEN register, an IRQ will be triggered. Writing '1' to these bit fields in the CAU_ERR_STATUS register will reset the error flags and hence the IRQ signal as well.



- **Event A:** ADC channel A is enabled via the ADC_EN signal in the CAU_ADCA register. However, the ADC won't be enabled until the CAU module is enabled first. Once CAU is enabled and ADC_EN is high, the ADC will enter its power-up period. After the power-up period is complete, the PWRD_UP[0] bit is set to indicate that ADC channel A is powered up and ready to receive triggers.
- **Event B:** The CAU is disabled while ADC_EN is still enabled. The ADC will be disabled but the CAU_ADCA configuration is retained. The PWRD_UP[0] bit is reset indicating that the ADC is disabled.
- **Event C:** The CAU module is enabled once again. Since there were no changes in the CAU_ADCA configuration and ADC_EN is still high, the ADC channel will be enabled once again, and it will enter the power-up period right away.
- **Event D:** ADC channel A is disabled via the CAU_ADCA register while CAU is still enabled. The ADC will be disabled right away and the PWRD_UP[0] bit is reset indicating that the ADC is disabled.
- **Event E:** A trigger is received while the ADC is still in its power-up period. The trigger is skipped and CTRIG_SKIP[0] and TRIG_PWRUP bits are set in the CAU_ERR_STATUS register to indicate that a trigger has been skipped because ADC channel A was still in its power-up period. An IRQ is generated if interrupt generation is enabled via the CAU_ERR_INTEN register.

Notes:
(1) PWRD_UP<2:0> signal is found in the CAU_PWRUP_STATUS register.
(2) t_{pwrup} is equal to 200 clock cycles.
(3) Signal is reset by software by writing '1' to the corresponding bits in CAU_ERR_STATUS registers. The IRQ signal follows these signals and hence will be reset too.

Figure 1-7: CAU behavior – example 4

1.6 DMA Transfer

It is possible to transfer the results of the ADC conversion (CAU_RESULTx or CAU_OFFSETx) by CAU to a specific memory location using DMA. This can minimise software effort as it is possible to record long array of data capture before analysing it. When a conversion is completed , a DMA request is triggered. When the total number of conversions are equal to the number of DMA request defined by NUM_ACQ in CAU_DMA_CONFIG register, a DMA transfer request will take place to transfer the data from DAU_RESULTx or DAU_OFFSETx registers to a desired memory destination defined by DMA[1]. CAU_DMA_EN in CAU_DMA _CONFIG register defines which one of the three ADC conversions should contribute to the total number of DMA request. For example if CAU_DMA_EN is set to 110_b , then the conversion complete for ADC_C and ADC_B will contribute to the total number of DMA request. In this case, if NUM_ACQ is set to 2, then the DMA transfer will take place once the conversions are completed for ADC_C and ADC_B. It is important that NUM_ACQ to be set to a value smaller or equal to the number of ADC channels contributing to the DMA request which is defined by CAU_DMA_EN or otherwise no DMA trasnfer will take place. For example if DMASLT_EN is set to 011_b, then the DMA request will be sent for ADC_A and ADC_B after conversion completion adding to a total of two DMA request. In this case, if NUM_ACQ is set to a value greater than 2, no DMA transfer will take place, because the total number of DMA request are done by only two ADC channels and hence it never exceeds 2 DMA requests. Upon completion of a DMA transfer, the pending requests reset and the next DMA transfer occurs once the next set of conversions occur.

Note: (1) The DMA module needs to be configured propaprly to define the starting and destination address for data to be sent correctly from CAU register to a desired memory address. (Refer to DMA document to configure the settings accordingly)

1.7 Register Set

The host can access the module's register set via the AMBA APB interface. Registers are word aligned. Reserved and unimplemented bits are read as zero.

Each ADC channel has its corresponding set of configuration registers. The module also features registers to control and read out all interrupt sources.

The register map of the CAU consists of 27 16-bit registers. Table 1 summarizes the register addresses, access types and reset values. The starting address of the CAU module is 0x40000000.

Base Address: 0x40000000

Table 1: Register map of the CAU module

Register Name	Access Type	Address Offset	Reset Value
CAU_CONTROL	R/W	0x00	0x0000
CAU_ADCA	R/W	0x04	0x0000
CAU_ADCB	R/W	0x08	0x0000
CAU_ADCC	R/W	0x0C	0x0000
CAU_TRIGA	R/W	0x10	0x0002
CAU_TRIGB	R/W	0x14	0x0002
CAU_TRIGC	R/W	0x18	0x0002
CAU_SWTRIG	R/W	0x1C	0x0000
CAU_CALIBRATIONA	R/W	0x20	0x0000
CAU_CALIBRATIONB	R/W	0x24	0x0000
CAU_CALIBRATIONC	R/W	0x28	0x0000
CAU_SWTRIG_CAL	R/W	0x2C	0x0000
CAU_IIRA	R/W	0x30	0x0000
CAU_IIRB	R/W	0x34	0x0000
CAU_IIRC	R/W	0x38	0x0000
CAU_ERR_STATUS	R/1C	0x3C	0x0000
CAU_ERR_INTEN	R/W	0x40	0x0000
CAU_PWRUP_STATUS	R	0x44	0x0000
CAU_CONV_STATUS	R/1C	0x48	0x0000
CAU_CONV_INTEN	R/W	0x4C	0x0000
CAU_OFFSETA	R	0x50	0x0000
CAU_OFFSETB	R	0x54	0x0000
CAU_OFFSETC	R	0x58	0x0000
CAU_RESULTA	R	0x5C	0x0000
CAU_RESULTB	R	0x60	0x0000
CAU_RESULTC	R	0x64	0x0000
CAU_DMA_CONFIG	R/W	0x68	0x0001

CAU_CONTROL

Reset value = 0x0000 RegisterAddress: 0x40000000

This register disables or enables the whole CAU System.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-	-	-	-	-	-	-	-
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
-	-	-	-	-	-	-	CAU_EN
U	U	U	U	U	U	U	R/W

Legend:	
R: Read	U: Unimplemented bit, read as 0
W: Write	Res: Reserved bit
1C: Write 1 to clear. Writing 0 has no effect	

Field	Bits	Description
CAU_EN	0	Master enable bit for the CAU module 0: Disabled (default) 1: Enabled

CAU_ADCx

Reset value = 0x0000
There are three ADCs in CAU module and x can represent one of three ADCs A, B or C.

Register Address CAU_ADCA: 0x40000004
Register Address CAU_ADCB: 0x40000008
Register Address CAU_ADCC: 0x4000000C

There are in total three ADCs for each phase, ADCA, ADCB and ADCC. The settings for each of these ADCs are independent. The settings in these registers define the conversion speed, the overwrite setting and if the corresponding ADC channel should be enabled. Also the settings in this reigster define if a conversion should be calibrated atuomatically or not.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-	-	-	-	-	-	-	-
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
-	-	-	OW_SEL	ADC_EN	CACNV_SPEED[1:0]		CM
U	U	U	R/W	R/W	R/W	R/W	R/W

Legend:
R: Read
W: Write
1C: Write 1 to clear. Writing 0 has no effect
U: Unimplemented bit, read as 0
Res: Reserved bit

Field	Bits	Description
ACM	0	**Automatic Calibration Mode** When enabled, the automatic calibration function is activated. The zero-current offset-voltage value is removed automatically from the measured voltage during conversions. The result is stored in the CAU_RESULTx register. 0: Disabled (default) 1: Enabled
CACNV_SPEED[1:0]	2:1	**Conversion Speed Selection.** 00_b: 1.0 µs (default) 01_b: 1.5 µs 10_b: 2.0 µs 11_b: 2.5 µs
ADC_EN	3	**ADC enable bit.** 0: Disabled (default) 1: Enabled
OW_SEL	4	**Overwrite Selection Bit.** Selects whether a new conversion will automatically overwrite the previous conversion stored in the CAU_RESULTx register if it has not been read yet. 0: Overwrite not permitted (default) 1: Overwrite permitted

CAU_TRIGx

Reset value = 0x0002
There are three ADCs in CAU module and x can represent one of three ADCs A, B or C.

Register Address CAU_TRIGA: 0x40000010
Register Address CAU_TRIGB: 0x40000014
Register Address CAU_TRIGC: 0x40000018

This register defines the track and hold period, the source selection for the normal trigger signal and the setting for the trigger edge detection.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-	-	-	-	-	-	-	CTESEL[1]
U	U	U	U	U	U	U	R/W

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
NTESEL[0]	NTSSEL[3:0]				CATH_PRD[2:0]		
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend:
R: Read
W: Write
1C: Write 1 to clear. Writing 0 has no effect
U: Unimplemented bit, read as 0
Res: Reserved bit

Field	Bits	Description
CATH_PRD[2:0]	2:0	**CAU Track and Hold Period** 000_b: 0 µs [1] 001_b: 0.5 µs 010_b: 1.0 µs (default) 011_b: 1.5 µs 100_b: 2.0 µs 101_b: 3.0 µs 110_b: 4.0 µs 111_b: 5.0 µs
NTSSEL[3:0]	6:3	**Normal Trigger Source Selection** 0000_b: ADCxNT [2] (default) 0001_b: sw_trig 0010_b: T1_Trigger [3] 0011_b: T2_Trigger [3] 0100_b: T3_Trigger [3] 0101_b: T4_Trigger [3] 0110_b: PD0 [4] 0111_b: PD1 [4] 1000_b: PD2 [4] 1001_b: PD3 [4] 1010_b: PD4 [4] 1011_b: PD5 [4] 1100_b: PD6 [4] 1101_b: PD7 [4] 1110_b: Reserved 1111_b: Reserved
NTESEL[1:0]	8:7	**Normal Trigger Edge Selection** 00_b: Rising edge (default) 01_b: Falling edge 10_b: Rising or falling edge 11_b: Rising edge
	Notes	(1) The T/H period value of 0 µs has been added due to the possibility of having some standby time between one conversion and the following. Meaning that no T/H period time is needed for the ADC. (2) Input from Pulse Width Modulation Unit (PWMU) (3) Input from general purpose timer unit (GTU) (4) External Input source connected to GPIO. The GPIO port needs to be configured as digital input in GPIO module. Refer to GPIO documentation for setting up GPIO as input.

CAU_SWTRIG

Reset value = 0x0000 Register Address: 0x4000001C

This register generates the normal trigger source when NTSSEL is set to sw_trig.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-	-	-	-	-	-	-	-
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
-	-	-	-	-	SW_TRIG_EN[2:0]		
U	U	U	U	U	R/W	R/W	R/W

Legend:
R: Read
W: Write
1C: Write 1 to clear. Writing 0 has no effect
U: Unimplemented bit, read as 0
Res: Reserved bit

Field	Bits	Description
SW_TRIG_EN[2:0]	2:0	**Software Normal Trigger Generation** This register will generate the normal trigger. SW_TRIG must be selected as the trigger source ('NTSSEL=0001') in the CAU_TRIGx register. 000_b: All software triggers disabled (default) 001_b: Set Normal Trigger High for ADC_A 010_b: Set Normal Trigger High for ADC_B 011_b: Set Normal Trigger High for ADC_B and ADC_A 100_b: Set Normal Trigger High for ADC_C 101_b: Set Normal Trigger High for ADC_C and ADC_A 110_b: Set Normal Trigger High for ADC_C and ADC_B 111_b: Set Normal Trigger High for all three ADC_C, ADC_B and ADC_A

CAU_CALIBRATIONx

Reset value = 0x0000
There are three ADCs in CAU module and x can represent one of three ADCs A, B or C.

Register Address CALIBRATIONA: 0x40000020
Register Address CALIBRATIONB: 0x40000024
Register Address CALIBRATIONC: 0x40000028

This register defines the source selection for the calibration trigger signal and the setting for the calibration trigger edge detection.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-	-	-	-	-	-	-	-
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
-	-	CTESEL[1:0]		CTSSEL[3:0]			
U	U	R/W	R/W	R/W	R/W	R/W	R/W

Legend:
R: Read
W: Write
1C: Write 1 to clear. Writing 0 has no effect
U: Unimplemented bit, read as 0
Res: Reserved bit

Field	Bits	Description
CTSSEL[3:0]	3:0	**Calibration Trigger Source Selection.** 0000_b: ADCxCT(1) (default) 0001_b: sw_trig_cal 0010_b: T1_trigger(2) 0011_b: T2_trigger (2) 0100_b: T3_trigger (2) 0101_b: T4_trigger (2) 0110_b: PD0 (3) 0111_b: PD1 (3) 1000_b: PD2 (3) 1001_b: PD3 (3) 1010_b: PD4 (3) 1011_b: PD5 (3) 1100_b: PD6 (3) 1101_b: PD7 (3) 1110_b: Reserved 1111_b: Reserved
CTESEL[1:0]	5:4	Calibration trigger edge selection. 00_b: Rising edge (default) 01_b: Falling edge 10_b: Rising or falling edge 11_b: Rising edge (1) Input from Pulse Width Modulation Unit (PWMU) (2) Input from general purpose timer unit (GTU) (3) External Input source connected to GPIO. The GPIO port needs to be configured as input in GPIO module. Refer to GPIO documentation for setting up required GPIO port as input.

CAU_SWTRIG_CAL

Reset value = 0x0000 Register Address: 0x4000002C

This register generates the calibration trigger source when CTSSEL is set to sw_cal_trig.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-	-	-	-	-	-	-	-
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
-	-	-	-	-	SW_CAL_EN[2:0]		
U	U	U	U	U	R/W	R/W	R/W

Legend:	
R: Read	U: Unimplemented bit, read as 0
W: Write	Res: Reserved bit
1C: Write 1 to clear. Writing 0 has no effect	

Field	Bits	Description
SW_CAL_EN[2:0]	2:0	**Software Calibration Trigger Generation** This register will generate the calibration trigger. SW_TRIG_CAL must be selected as the trigger source (‘CTSSEL=0001’) in the CAU_CALIBRATIONx register. 000_b: All software triggers disabled (default) 001_b: Set Calibration Trigger High for ADC_A 010_b: Set Calibration Trigger High for ADC_B 011_b: Set Calibration Trigger High for ADC_B and ADC_A 100_b: Set Calibration Trigger High for ADC_C 101_b: Set Calibration Trigger High for ADC_C and ADC_A 110_b: Set Calibration Trigger High for ADC_C and ADC_B 111_b: Set Calibration Trigger High for all three ADC_C, ADC_B and ADC_A

CAU_IIRx

Reset value = 0x0000
There are three ADCs in CAU module and x can represent one of three ADCs A, B or C.

Register Address IIRA: 0x40000030
Register Address IIRB: 0x40000034
Register Address IIRC: 0x40000038

This register set the enable setting and coefficient factor for IIR filter.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-	-	-	-	-	-	-	-
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
-	-	-	-	-	IIR_EN	CF_COE[1:0]	
U	U	U	U	U	R/W	R/W	R/W

Legend:
R: Read
W: Write
1C: Write 1 to clear. Writing 0 has no effect
U: Unimplemented bit, read as 0
Res: Reserved bit

Field	Bits	Description
CF_COE[1:0]	1:0	**CAU Filter Coefficient** The value indicates the dividing factor for the IIR filter coefficients. The dividing factor is given by 2^n. Check this ADC Offset Measurement section for further information. 00_b: n=0 01_b: n=1 10_b: n=2 11_b: n=3
IIR_EN	2	**IIR filter enable bit.** The IIR filter is used only during the calculation of the offset value. 0: Disabled (default) 1: Enabled

CAU_ERR_STATUS

Reset value = 0x0000 Register Address: 0x4000003C

This register contains the information about over-run , out of range, register configuration error , trigger skip , power up trigger error and normal and calibration trigger overlap error flags.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
TRIG_ERR[2:0]			TRIG_PWRUP	CTRIG_SKIP[2:0]			RCONF_ERR[2]
R/1C	R/1C	R/1C	R/1C	R/1C	R/1C	R/1C	R/1C

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
RCONF_ERR[1:0]		OOR[2:0]			OVR[2:0]		
R/1C	R/1C	R/1C	R/1C	R/1C	R/1C	R/1C	R/1C

Legend:
R: Read
W: Write
1C: Write 1 to clear. Writing 0 has no effect
U: Unimplemented bit, read as 0
Res: Reserved bit

Field	Bits	Description
OVR[2:0]	2:0	**OVER-RUN** This flag is set if a new conversion is complete but the old information in the CAU_RESULTx register has not been read yet. The old result may or may not be overwritten, depending on the overwrite setting in the CAU_ADCx register. There are three bits each correspond to an ADC channel. Writing ‘1’ to this field clears the flag. Interrupt generation must be enabled by configuring the CAU_ERR_INTEN register to generate an IRQ. 000_b: No Over-Run Detected 001_b: Over-Run Detected for ADC_A Conversion 010_b: Over-Run Detected for ADC_B Conversion 011_b: Over-Run Detected for ADC_B and ADC_A Conversion 100_b: Over-Run Detected for ADC_C Conversion 101_b: Over-Run Detected for ADC_C and ADC_A Conversion 110_b: Over-Run Detected for ADC_C and ADC_B Conversion 111_b: Over-Run Detected for all three ADC_C, ADC_B and ADC_A Conversion
OOR[2:0]	5:3	**Out-of-Range flag** Indicates any out-of-range detection (overflow or underflow) between the non-calibrated result and the offset value. There are three bits each correspond to an ADC channel. Writing ‘1’ to this field clears the flag. Interrupt generation must be enabled by configuring CAU_ERR_INTEN register to generate an IRQ. 000_b: No out of range Detected for a conversion 001_b: out of range Detected for ADC_A Conversion 010_b: out of range Detected for ADC_B Conversion 011_b: out of range Detected for ADC_B and ADC_A Conversion 100_b: out of range Detected for ADC_C Conversion 101_b: out of range Detected for ADC_C and ADC_A Conversion 110_b: out of range Detected for ADC_C and ADC_B Conversion 111_b: out of range Detected for all three ADC_C, ADC_B and ADC_A Conversion

RCONF_ERR[2:0]	8:6	**Configuration error flag.** This flag is set when an attempt to modify the contents of the CAU_ADCx, CAU_TRIGx or CAU_CALIBRATIONx registers takes places while a conversion is ongoing[1]. Writing '1' to this field clears the flag. Interrupt generation must be enabled by configuring the CAU_ERR_INTEN register to generate an IRQ. 000_b: No Register Configuration Error Detected 001_b: Register Configuration Error Detected for ADC_A 010_b: Register Configuration Error Detected for ADC_B 011_b: Register Configuration Error Detected for ADC_B and ADC_A 100_b: Register Configuration Error Detected for ADC_C 101_b: Register Configuration Error Detected for ADC_C and ADC_A 110_b: Register Configuration Error Detected for ADC_C and ADC_B 111_b: Register Configuration Error Detected for all three ADC_C, ADC_B and ADC_A
CTRIG_SKIP[2:0]	11:9	**CAU Trigger Skip flag.** This flag indicates that a trigger has been ignored either because a conversion was ongoing, or the ADC was still powering up when the trigger was received. Each bit corresponds to an ADC channel. Writing '1' to this field clears the flag. Interrupt generation must be enabled by configuring the CAU_ERR_INTEN register to generate an IRQ. 000_b: No Trigger Skip Detected 001_b: Trigger Skip Detected for ADC_A 010_b: Trigger Skip Detected for ADC_B 011_b: Trigger Skip Detected for ADC_B and ADC_A 100_b: Trigger Skip Detected for ADC_C 101_b: Trigger Skip Detected for ADC_C and ADC_A 110_b: Trigger Skip Detected for ADC_C and ADC_B 111_b: Trigger Skip Detected for all three ADC_C, ADC_B and ADC_A
TRIG_PWRUP[2]	12	**Trigger received during ADC power-up flag** This flag is asserted when any ADC channel receives a normal or calibration trigger before it has completed its power-up cycle. The trigger will be ignored in this case. ADC power-up information is available in the CAU_PWRUP_STATUS register. Writing '1' to this field clears the fault flag. Interrupt generation must be enabled by configuring the CAU_ERR_INTEN register to generate an IRQ. 0: No triggers received during power-up error 1: Trigger received during power-up error
TRIG_ERR[2:0]	15:13	**Trigger overlap error flag** This flag indicates that both normal and calibration triggers have been set at the same time for a given ADC channel. In this case, the normal trigger is prioritized and the calibration trigger is ignored. Writing '1' to this field clears the flag. Interrupt generation must be enabled in the CAU_ERR_INTEN register to generate an IRQ. 000_b: No Trigger Skip Detected 001_b: Trigger Overlap Error Detected for ADC_A 010_b: Trigger Overlap Error Detected for ADC_B 011_b: Trigger Overlap Error Detected for ADC_B and ADC_A 100_b: Trigger Overlap Error Detected for ADC_C 101_b: Trigger Overlap Error Detected for ADC_C and ADC_A 110_b: Trigger Overlap Error Detected for ADC_C and ADC_B 111_b: Trigger Overlap Error Detected for all three ADC_C, ADC_B and ADC_A

Notes

(1) Except for disabling the ADC channel via the ADC_EN field in the CAU_ADCx register.
(2) This bit along with the information in the CTRIG_SKIP[2:0] field will indicate which ADC channel caused the fault.

CAU_ERR_INTEN

Reset value = 0x0000 Register Address: 0x40000040

This register will enable the corresponding interrupt generation for errors reported in CAU_ERR_STATUS register.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
TRIG_ERR_EN[2:0]			TRIG_PWRUP_EN	CTRIG_SKIP_EN[2:0]			RCONF_ERR_EN[2]
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
RCONF_ERR_EN[1:0]		OOR_EN[2:0]			OVR_EN[2:0]		
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend:	
R: Read	U: Unimplemented bit, read as 0
W: Write	Res: Reserved bit
1C: Write 1 to clear. Writing 0 has no effect	

Field	Bits	Description
OVR_EN[2:0]	2:0	**Overrun Interrupt enable** This field enables or disables interrupts caused by an OVR error. The error information will be reported in the CAU_ERR_STATUS register even if the interrupt is disabled. Only the interrupt generation is affected by changing this field. 000_b: Interrupt Generation is disabled for OVR flag 001_b: Interrupt Generation is enabled for OVR flag in ADC_A 010_b: Interrupt Generation is enabled for OVR flag in ADC_B 011_b: Interrupt Generation is enabled for OVR flag in ADC_B and ADC_A 100_b: Interrupt Generation is enabled for OVR flag in ADC_C 101_b: Interrupt Generation is enabled for OVR flag in ADC_C and ADC_A 110_b: Interrupt Generation is enabled for OVR flag in ADC_C and ADC_B 111_b: Interrupt Generation is enabled for OVR flag in ADC_C, ADC_B and ADC_A
OOR_EN[2:0]	5:3	**Out-of-range Interrupt enable** This field enables or disables interrupts caused by an OOR error. The error information will be reported in the CAU_ERR_STATUS register even if the interrupt is disabled. Only the interrupt generation is affected by changing this field. 000_b: Interrupt Generation is disabled for OOR flag 001_b: Interrupt Generation is enabled for OOR flag in ADC_A 010_b: Interrupt Generation is enabled for OOR flag in ADC_B 011_b: Interrupt Generation is enabled for OOR flag in ADC_B and ADC_A 100_b: Interrupt Generation is enabled for OOR flag in ADC_C 101_b: Interrupt Generation is enabled for OOR flag in ADC_C and ADC_A 110_b: Interrupt Generation is enabled for OOR flag in ADC_C and ADC_B 111_b: Interrupt Generation is enabled for OOR flag in ADC_C, ADC_B and ADC_A

RCONF_ERR_EN[2:0]	8:6	**Register Configuration Error Interrupt Enable** This field enables or disables interrupts caused by a RCONF_ERR error. The error information will be reported in the CAU_ERR_STATUS register even if the interrupt is disabled. Only the interrupt generation is affected by changing this field. 000_b: Interrupt Generation is disabled for RCONF_ERR flag 001_b: Interrupt Generation is enabled for RCONF_ERR flag in ADC_A 010_b: Interrupt Generation is enabled for RCONF_ERR flag in ADC_B 011_b: Interrupt Generation is enabled for RCONF_ERR flag in ADC_B and ADC_A 100_b: Interrupt Generation is enabled for RCONF_ERR flag in ADC_C 101_b: Interrupt Generation is enabled for RCONF_ERR flag in ADC_C and ADC_A 110_b: Interrupt Generation is enabled for RCONF_ERR flag in ADC_C and ADC_B 111_b: Interrupt Generation is enabled for RCONF_ERR flag in ADC_C, ADC_B and ADC_A
CTRIG_SKIP_EN[2:0]	11:9	**Trigger Skip Interrupt Enable** This field enables or disables interrupts caused by a CTRIG_SKIP error. The error information will be reported in the CAU_ERR_STATUS register even if the interrupt is disabled. Only the interrupt generation is affected by changing this field. 000_b: Interrupt Generation is disabled for CTRIG_SKIP flag 001_b: Interrupt Generation is enabled for CTRIG_SKIP flag in ADC_A 010_b: Interrupt Generation is enabled for CTRIG_SKIP flag in ADC_B 011_b: Interrupt Generation is enabled for CTRIG_SKIP flag in ADC_B and ADC_A 100_b: Interrupt Generation is enabled for CTRIG_SKIP flag in ADC_C 101_b: Interrupt Generation is enabled for CTRIG_SKIP flag in ADC_C and ADC_A 110_b: Interrupt Generation is enabled for CTRIG_SKIP flag in ADC_C and ADC_B 111_b: Interrupt Generation is enabled for CTRIG_SKIP flag in ADC_C, ADC_B and ADC_A
TRIG_PWRUP_EN	12	**Trigger Power-up Error Interrupt Enable** This field enables or disables interrupts caused by a TRIG_PWRUP error. The error information will be reported in the CAU_ERR_STATUS register even if the interrupt is disabled. Only the interrupt generation is affected by changing this field. 0: Trigger during power-up error interrupt disabled (default) 1: Trigger during power-up error interrupt enabled
TRIG_ERR_EN[2:0]	15:13	**Trigger Overlap Error Interrupt Enable** This field enables or disables interrupts caused by a TRIG_ERR. The error information will be reported in the CAU_ERR_STATUS register even if the interrupt is disabled. Only the interrupt generation is affected by changing this field. 000_b: Interrupt Generation is disabled for TRIG_ERR flag 001_b: Interrupt Generation is enabled for TRIG_ERR flag in ADC_A 010_b: Interrupt Generation is enabled for TRIG_ERR flag in ADC_B 011_b: Interrupt Generation is enabled for TRIG_ERR flag in ADC_B and ADC_A 100_b: Interrupt Generation is enabled for TRIG_ERR flag in ADC_C 101_b: Interrupt Generation is enabled for TRIG_ERR flag in ADC_C and ADC_A 110_b: Interrupt Generation is enabled for TRIG_ERR flag in ADC_C and ADC_B 111_b: Interrupt Generation is enabled for TRIG_ERR flag in ADC_C, ADC_B and ADC_A

CAU_PWRUP_STATUS

Reset value = 0x0000 Register Address: 0x40000044

This register indicates the ADC power-up status.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-	-	-	-	-	-	-	-
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
-	-	-	-	-	PWRD_UP[2:0]		
U	U	U	U	U	R	R	R

Legend:
R: Read
W: Write
1C: Write 1 to clear. Writing 0 has no effect
U: Unimplemented bit, read as 0
Res: Reserved bit

Field	Bits	Description
PWRD_UP [2:0]	2:0	**Power-up** This field indicates the ADCs power-up status. The value 1 indicates that the ADC channel is powered-up and is ready to perform a conversion. The value 0 indicates that the ADC channel is either disabled or still powering-up. This field shall be checked after enabling an ADC channel to know when the ADC has completed its power-up cycle. Each bit corresponds to an ADC channel in the following order: cba 000_b: All ADC channels are disabled or powering up. 001_b: ADC_A is powered up 010_b: ADC_B is powered up 011_b: ADC_A and ADC_B are powered up 100_b: ADC_C is powered up 101_b: ADC_A and ADC_C are powered up 110_b: ADC_B and ADC_C are powered up 111_b: All ADC channels are powered up

CAU_CONV_STATUS

Reset value = 0x0000 Register Address: 0x40000048

This register will provide a status report if a conversion has been completed by a normal or calibration trigger.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-	-	-	-	-	-	-	-
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
-	-	OCC[2:0]			CC[2:0]		
U	U	R/1C	R/1C	R/1C	R/1C	R/1C	R/1C

Legend:
R: Read
W: Write
1C: Write 1 to clear. Writing 0 has no effect
U: Unimplemented bit, read as 0
Res: Reserved bit

Field	Bits	Description
CC[2:0]	2:0	**Conversion Completed Flag** This flag is set when an ADC channel completes a conversion, triggered by a normal trigger, and the information is ready to be read from the CAU_RESULTx register. This flag will trigger an IRQ to the system NVIC if the corresponding 'CC_EN[2:0]' bits are set in the CAU_CONV_INTEN register. Writing '1' to this field clears the flag. 000_b: No conversion is completed by a normal trigger 001_b: A conversion is completed by a normal trigger for ADC_A 010_b: A conversion is completed by a normal trigger for ADC_B 011_b: A conversion is completed by a normal trigger for ADC_B and ADC_A 100_b: A conversion is completed by a normal trigger for ADC_C 101_b: A conversion is completed by a normal trigger for ADC_C and ADC_A 110_b: A conversion is completed by a normal trigger for ADC_C and ADC_B 111_b: A conversion is completed by a normal trigger for ADC_C, ADC_B and ADC_A
OCC[2:0]	5:3	**Offset Conversion Completed Flag** This flag is set when an ADC channel completes a calibration conversion (offset calculation), triggered by a calibration trigger, and the information is ready to be read from the CAU_OFFSETx register. This flag will trigger an IRQ to the system NVIC if the corresponding 'OCC_EN[2:0]' bits are set in the CAU_CONV_INTEN register. Writing '1' to this field clears the flag. 000_b: No conversion is completed by a calibration trigger 001_b: A conversion is completed by a calibration trigger for ADC_A 010_b: A conversion is completed by a calibration trigger for ADC_B 011_b: A conversion is completed by a calibration trigger for ADC_B and ADC_A 100_b: A conversion is completed by a calibration trigger for ADC_C 101_b: A conversion is completed by a calibration trigger for ADC_C and ADC_A 110_b: A conversion is completed by a calibration trigger for ADC_C and ADC_B 111_b: A conversion is completed by a calibration trigger for ADC_C, ADC_B and ADC_A

CAU_CONV_INTEN

Reset value = 0x0000 Register Address: 0x4000004C

This register enables or disables the interrupt generation for the corresponding bits set in CAU_CONV_STATUS register.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-	-	-	-	-	-	-	-
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
-	-	OCC_EN[2:0]			CC_EN[2:0]		
U	U	R/W	R/W	R/W	R/W	R/W	R/W

Legend:
R: Read
W: Write
1C: Write 1 to clear. Writing 0 has no effect
U: Unimplemented bit, read as 0
Res: Reserved bit

Field	Bits	Description
CC_EN[2:0]	2:0	**Conversion completed Interrupt Enable** This field enables the interrupt generation for each ADC channel when a conversion is complete. Conversion completion will still be reported in the CAU_CONV_STATUS register even if this field is disabled. Only interrupt generation is affected by changing this field. 000_b: No Interrupt is enabled 001_b: Interrupt generation is enabled for CC status bit for ADC_A 010_b: Interrupt generation is enabled for CC status bit for ADC_B 011_b: Interrupt generation is enabled for CC status bit for ADC_B and ADC_A 100_b: Interrupt generation is enabled for CC status bit for ADC_C 101_b: Interrupt generation is enabled for CC status bit for ADC_C and ADC_A 110_b: Interrupt generation is enabled for CC status bit for ADC_C and ADC_B 111_b: Interrupt generation is enabled for CC status bit for ADC_C, ADC_B and ADC_A
OCC_EN[2:0]	5:3	**Offset Conversion Completed Interrupt Enable** This field enables the interrupt generation for each ADC channel when a calibration conversion (offset calculation) is complete. Calibration conversion completion will still be reported in the CAU_CONV_STATUS register even if this field is disabled. Only interrupt generation is affected by changing this field. 000_b: No Interrupt is enabled 001_b: Interrupt generation is enabled for OCC status bit for ADC_A 010_b: Interrupt generation is enabled for OCC status bit for ADC_B 011_b: Interrupt generation is enabled for OCC status bit for ADC_B and ADC_A 100_b: Interrupt generation is enabled for OCC status bit for ADC_C 101_b: Interrupt generation is enabled for OCC status bit for ADC_C and ADC_A 110_b: Interrupt generation is enabled for OCC status bit for ADC_C and ADC_B 111_b: Interrupt generation is enabled for OCC status bit for ADC_C, ADC_B and ADC_A

CAU_OFFSETx

Reset value = 0x0000
There are three ADCs in CAU module and x can represent one of three ADCs A, B or C.

Address Offset: CAU_OFFSETA: 0x40000050
Address Offset: CAU_OFFSETB: 0x40000054
Address Offset: CAU_OFFSETC: 0x40000058

These registers record the result of the ADC measurement across shunt resistor after a successful conversion initiated by a calibration trigger. The result is a 12-bit singed number. The two least significant bits are padded with 0 and the most two significant bits are used for sign extension.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
CALRES Sign Extension		CALRES[11:6]					
R	R	R	R	R	R	R	R

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
CALRES[5:0]						0	0
R	R	R	R	R	R	R	R

Legend:
R: Read
W: Write
1C: Write 1 to clear. Writing 0 has no effect
U: Unimplemented bit, read as 0
Res: Reserved bit

Field	Bits	Description
CALRES[11:0][1](S)	13:2	A signed number showing calculated reference offset voltage across shunt resistor after the execution of a calibration conversion for each ADC channel. Each ADC channel has its corresponding CAU_OFFSETx register.

$$V_{RSENSECAL}(V) = \frac{\textbf{CALRES}}{2048} \times \frac{1.2}{SAG}$$

Where SAG is the sense amplifier gain that can be modified by register 1 in GDU (refer to GDU documentation for further information).

For example when SAG is set to ‘5’, and CALRES[11:0] value set to ‘60’, then the voltage across sense resistor can be calculated as below:

$$V_{RSENSECAL} = \frac{60}{2048} \times \frac{1.2}{5} = 7.03mV$$

(1) To obtain CALRES value, read the corresponding CAU_OFFSETx register as a 16-bit signed number and shift that value by two bits to the right.

(2) If IIR filter is enabled, then the CALRES value will be the output of IIR filter (check ADC OFFSET measurement section). Otherwise, the CALRES value will be the direct voltage measurement across the sense resistor initiated by a calibration trigger.

CAU_RESULTx

Reset value = 0x0000
There are three ADCs in CAU module and x can represent one of three ADCs A, B or C.

Address Offset: CAU_RESULTA: 0x4000005C
Address Offset: CAU_RESULTB: 0x40000060
Address Offset: CAU_RESULTC: 0x40000064

These registers record the result of the ADC measurement across shunt resistor after a successful conversion initiated by a normal trigger. The result is a 12-bit singed number. The two least significant bits are padded with 0s and the most two significant bits are used for sign extension. If automatic calibration (ACM) is enabled, CALRES value will be the result of ADC measurement subtracted by offset value set in CAU_OFFSETx Register. Otherwise CALRES will be the direct result of ADC measurement.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
CALRES Sign Extension		NORMRES[11:6]					
R	R	R	R	R	R	R	R

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
NORMRES[5:0]						0	0
R	R	R	R	R	R	R	R

Legend:	
R: Read	U: Unimplemented bit, read as 0
W: Write	Res: Reserved bit
1C: Write 1 to clear. Writing 0 has no effect	

Field	Bits	Description
NORMRES[11:0][1](S)	13:2	A signed number showing calculated reference offset voltage across shunt resistor after the execution of a normal conversion for each ADC channel. Each ADC channel has its corresponding CAU_RESULTx register.

$$V_{RSENSE}(V) = \frac{\mathbf{NORMRES}}{2048} \times \frac{1.2}{SAG}$$

Where SAG is the sense amplifier gain that can be modified by register 1 in GDU (refer to GDU documentation for further information).

For example when SAG is set to ‘5’, and NORMRES[11:0] value set to ‘1800’, then the voltage across sense resistor can be calculated as below:

$$V_{RSENSECAL} = \frac{1800}{2048} \times \frac{1.2}{5} = 211mV$$

(1) To obtain NORMRES value, read the corresponding CAU_RESULTx register as a 16-bit signed number and shift it by two bits to the right.

(2) When automatic calibration (ACM) is enabled, NORMRES will be the result of direct measurement done by ADC after normal trigger subtracted by corresponding CAU_OFFSETx register. When automatic calibration is disabled, NORMRES will be the direct measurement done by ADC after normal trigger.

CAU_DMA_CONFIG

Reset value = 0x0001 Register Address: 0x40000068

This register will define the settings for DMA trasnfer.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-	-	-	-	-	-	-	-
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
-	-	-	DMA_EN[2:0]			NUM_ACQ[1:0]	
U	U	U	U	U	U	R/W	R/W

Legend:
R: Read
W: Write
1C: Write 1 to clear. Writing 0 has no effect
U: Unimplemented bit, read as 0
Res: Reserved bit

Field	Bits	Description
NUM_ACQ[1:0]	1:0	**Number of Acquisitions** Number of acquisitions to be accumulated before sending notification to the DMA controller. 00: 0 acquisitions 01: 1 acquistion (default) 10: 2 acquistions 11: 3 acquistions
DMA_EN[2:0]	4:2	**DMA Resquest Enable** DMA request enable per ADC channel. 000: DMA request is disabled for all ADC channels 001: DMA request is enabled for ADC_A 010: DMA request is enabled for ADC_B 011: DMA request is enabled for ADC_A and ADC_B 100: DMA request is enabled for ADC_C 101: DMA request is enabled for ADC_A and ADC_C 110: DMA request is enabled for ADC_B and ADC_C 111: DMA request is enabled for all ADC channels

March 22/2021	Initial Revision
Nov 11/2021	Added Revision Control Updated Logo
July 31/2024	Changed description from sample and hold to track and hold
Nov 11/2024	Updated the DMA section

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